



BIO262

Physiology

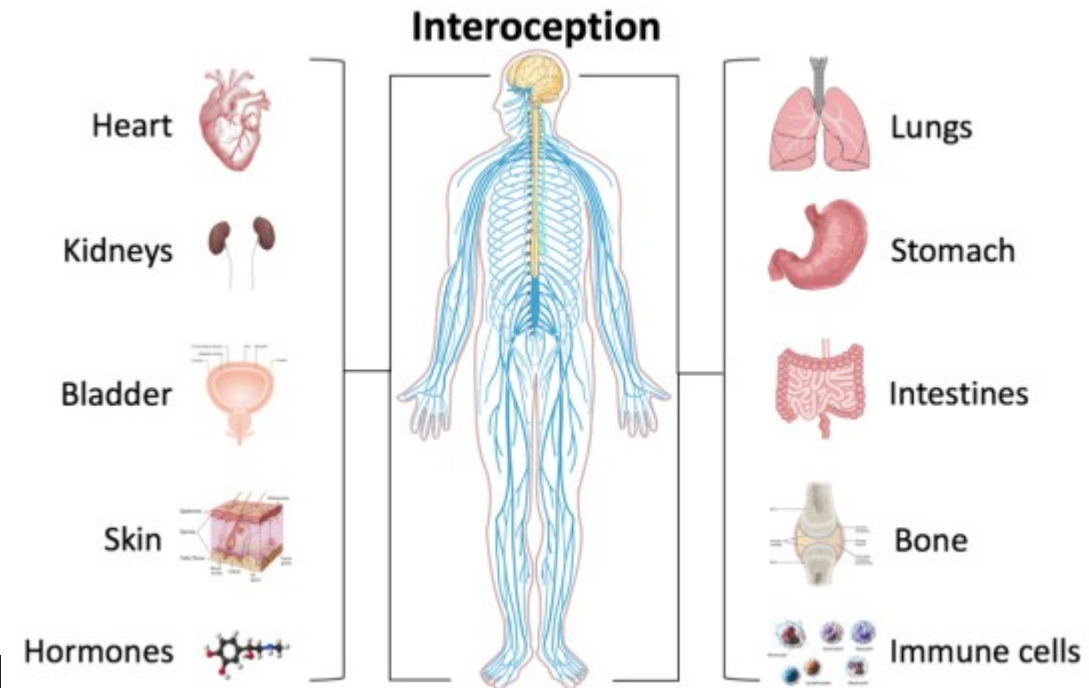
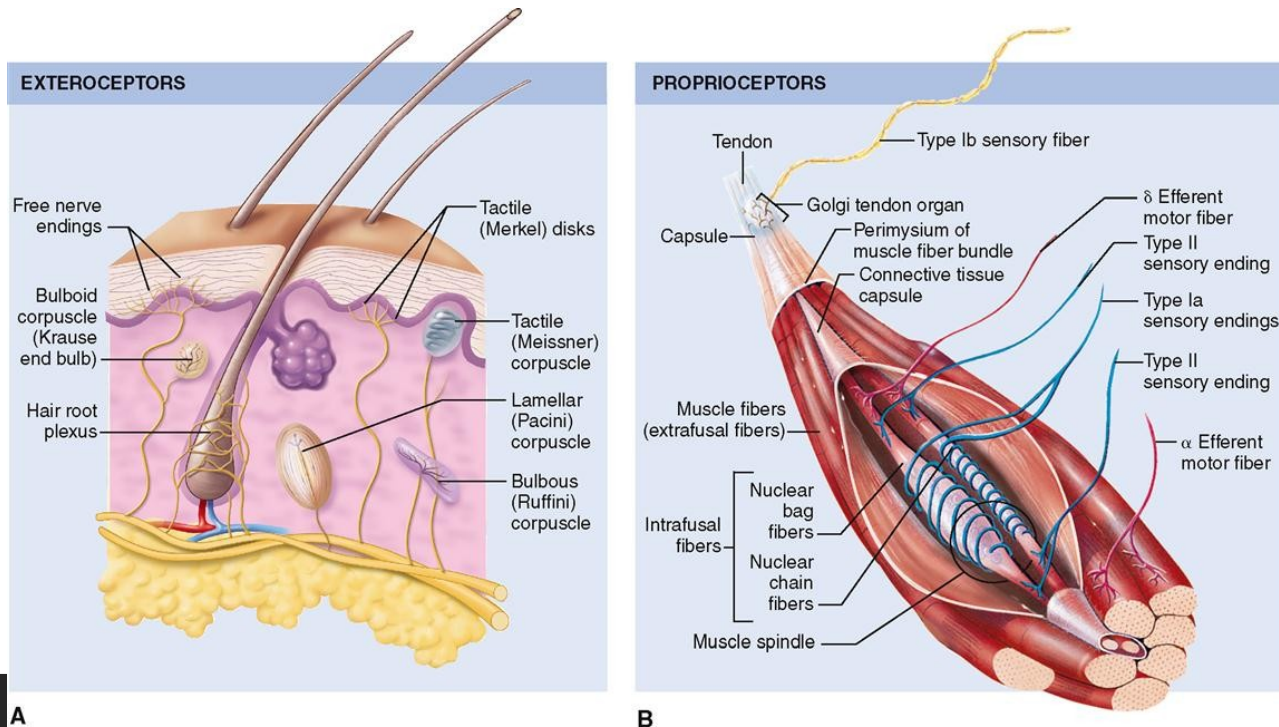
MODULE 8

RECEPTORS

HORMONES

Receptor class by location

- Exteroceptors – monitor the external environment
- Proprioceptors – monitor and maintain balance (joints/muscles)
- Interoceptors – monitor the internal environment (visceral organs)



Receptors by Stimulus Excitability

Nociceptors (protective) – PAIN

Thermoreceptors - Temperature

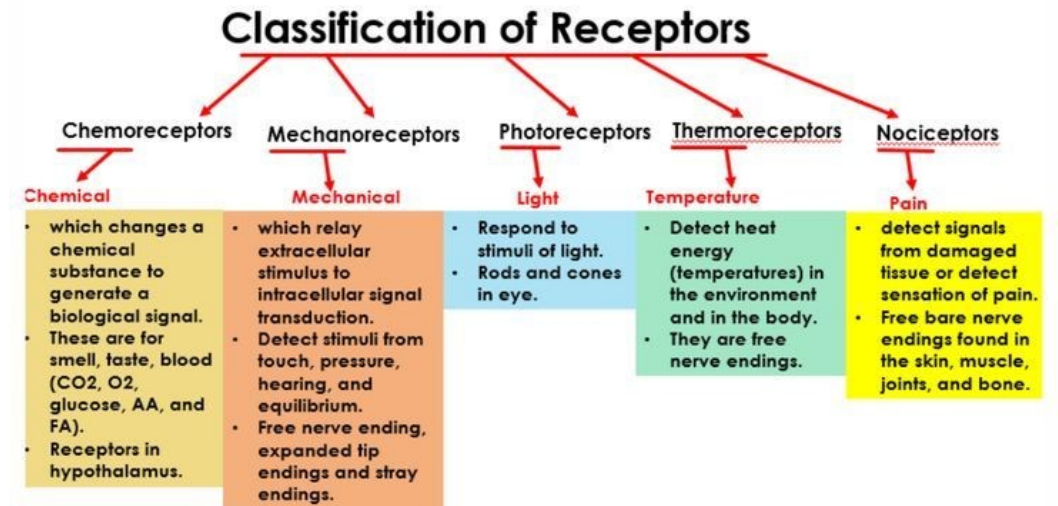
Chemoreceptors – Chemicals: O₂, CO₂, pH....

Mechanoreceptors – distortion of cell membrane – vibration, pressure.....

Baroreceptors (a specialized mechanoreceptor)
– internal organs, vessels – change in pressure.
Monitors BP/blood volume for example.

Photoreceptors – respond to light – eyes.

CLASSIFICATION OF RECEPTORS



Lipid soluble hormones enter the cell and bind to internal receptors – communicate with DNA.

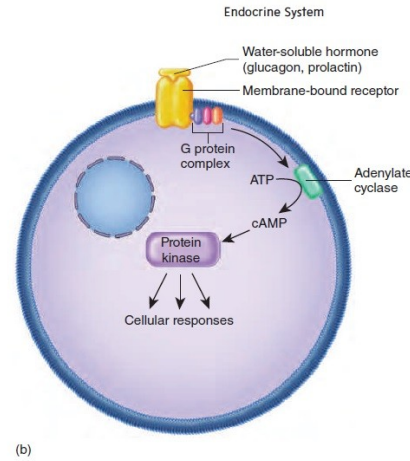
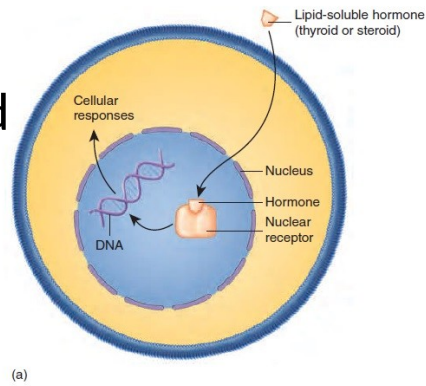
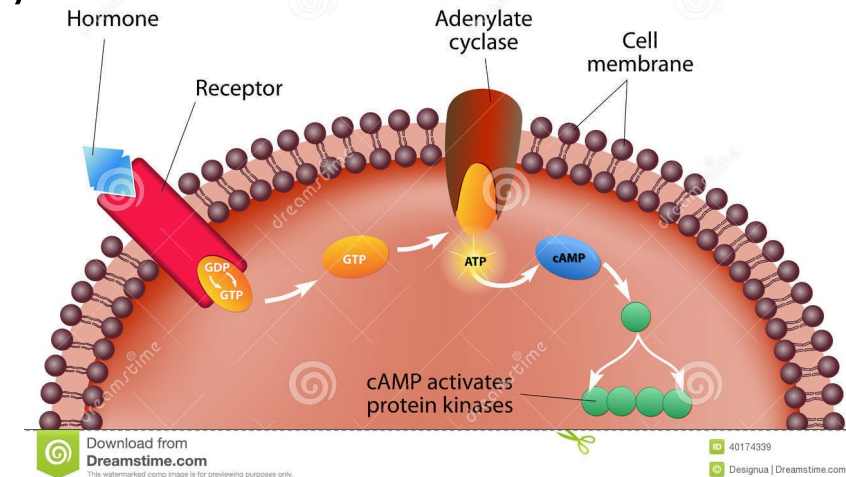


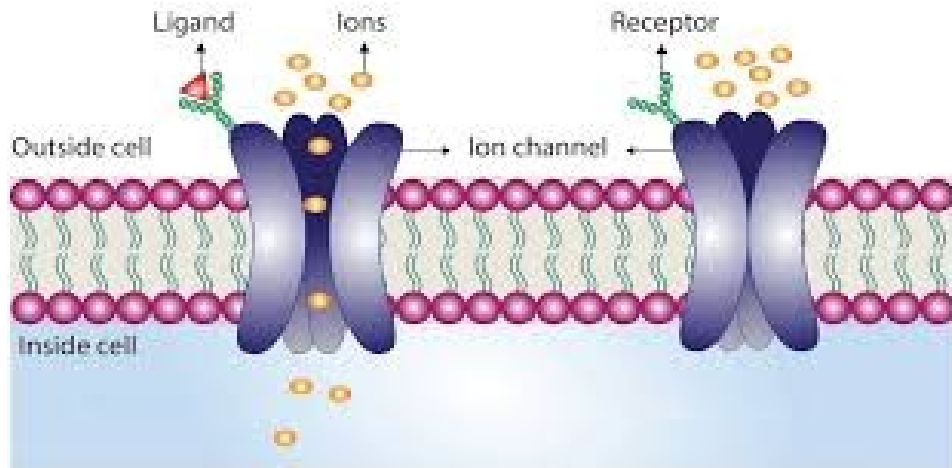
Figure 10.7 General Comparison of Nuclear and Membrane-Bound Receptors
 (a) Lipid-soluble hormones diffuse through the plasma membrane of its target cell and bind to a cytoplasmic receptor or a nuclear receptor. In the nucleus, the combination of the hormone and the receptor initiates protein synthesis, described later in this chapter. (b) Water-soluble hormones bind to the external portion of membrane-bound receptors, which are integral membrane proteins on its target cell.

Water soluble hormones cannot get through lipid tails in cell membrane. They bind to extracellular receptors.

MECHANISMS OF HORMONE ACTION



Some extracellular receptors are G-protein linked which activate intracellular G-proteins.



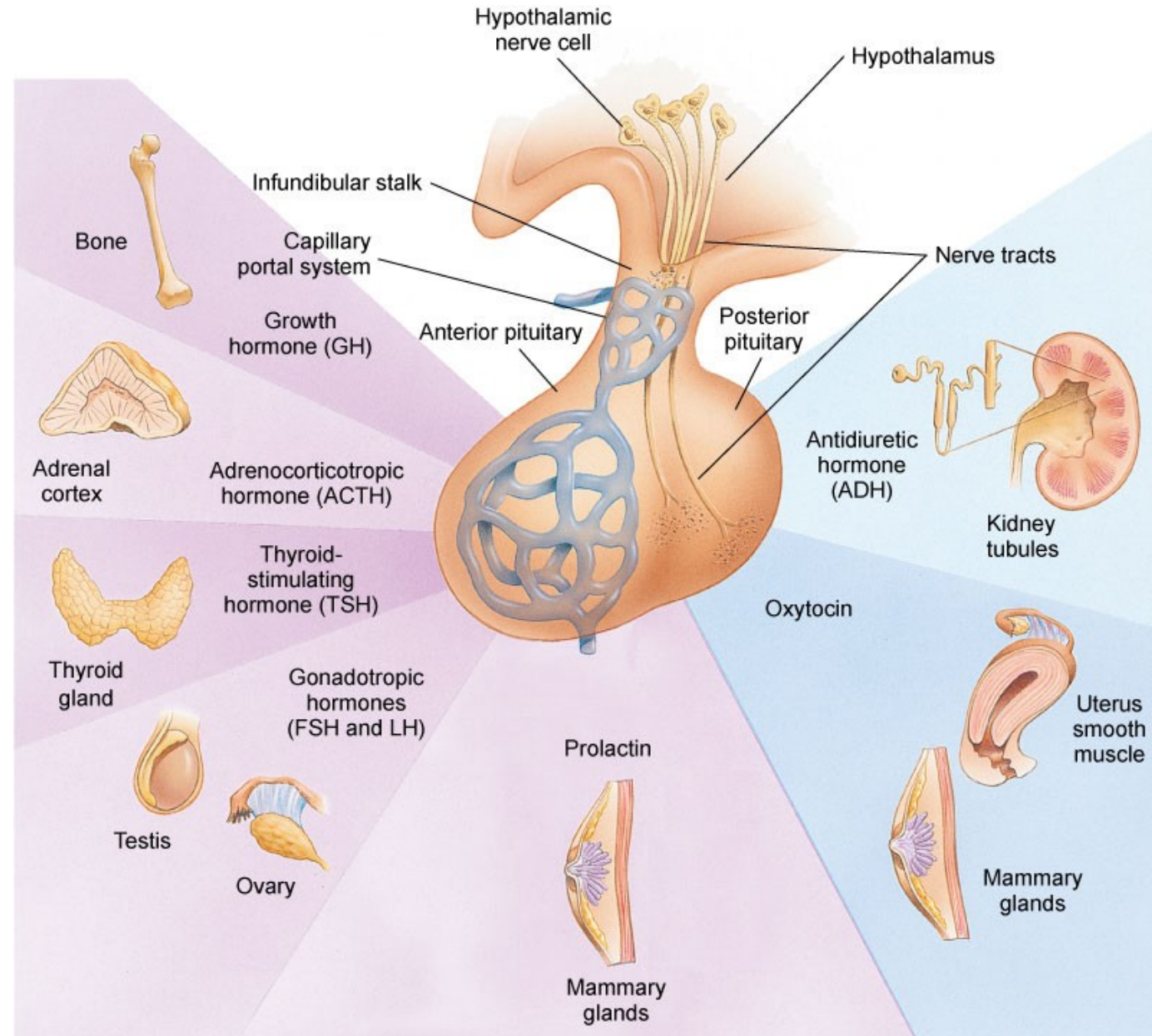
ION channel receptors – when ligand like neurotransmitter binds it opens an ion gate and ions move. Eg: Ach in muscle contractions

WHY do we need/have HORMONES?

- ✓ Alter transcription/translation – therefore changes protein/enzyme production
- ✓ Changes enzymes/channel shapes – turns on or off
- ✓ Alters cellular activity in various tissues at the same time
- ✓ Coordinates cell, tissue, organ activities – long-term!

Hypothalamus and the Pituitary Gland

Hypothalamus monitors many hormone levels. It then communicates with the pituitary gland which then releases hormones that target other body parts to release hormones OR pituitary hormones directly impact tissues.

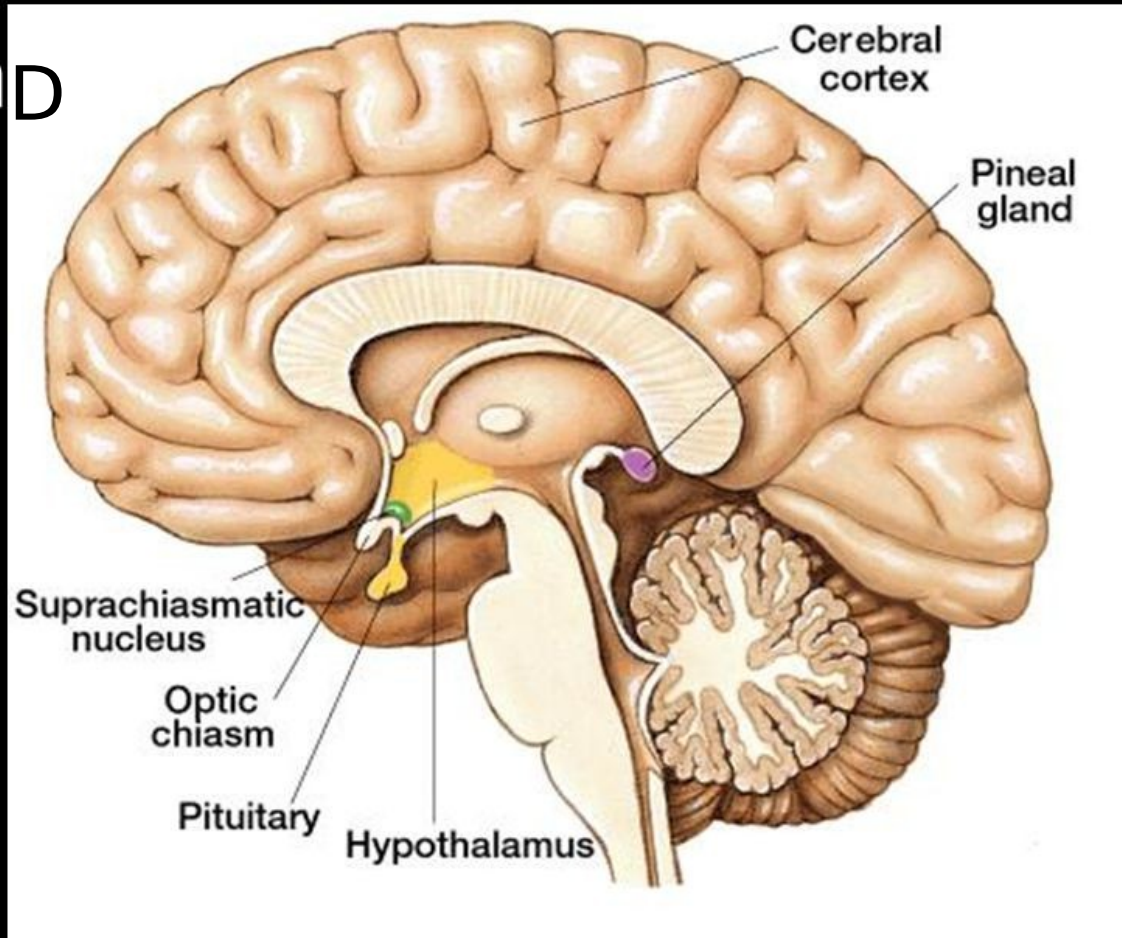


Pineal Gland: Melatonin - THINK SLEEP!!!

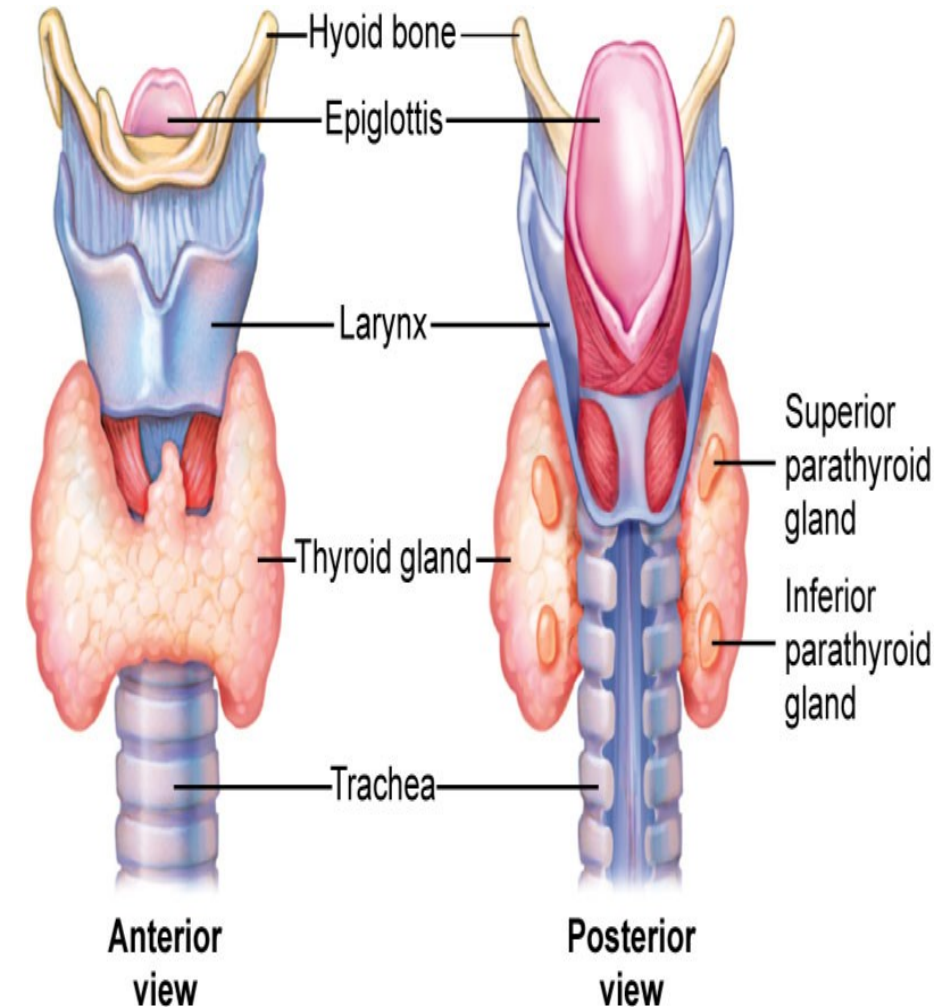
Light/dark control melatonin release – controls sleep/wake cycle (circadian rhythms)

The Pineal Gland

- Functions: the pineal gland releases hormones that control circadian rhythm and regulate certain reproductive hormones.
- Hormones:
 - Melatonin
 - Serotonin



- Thyroid stimulating hormone FROM ANT PIT due to TRH from Hypothalamus.
- TRH is released when T3/T4 blood levels are LOW.
- Target: thyroid
- Tells Thyroid to release T3 and T4 (Thyroxine)
 - Target of T3/T4 – all cells
 - Controls **metabolism**/energy

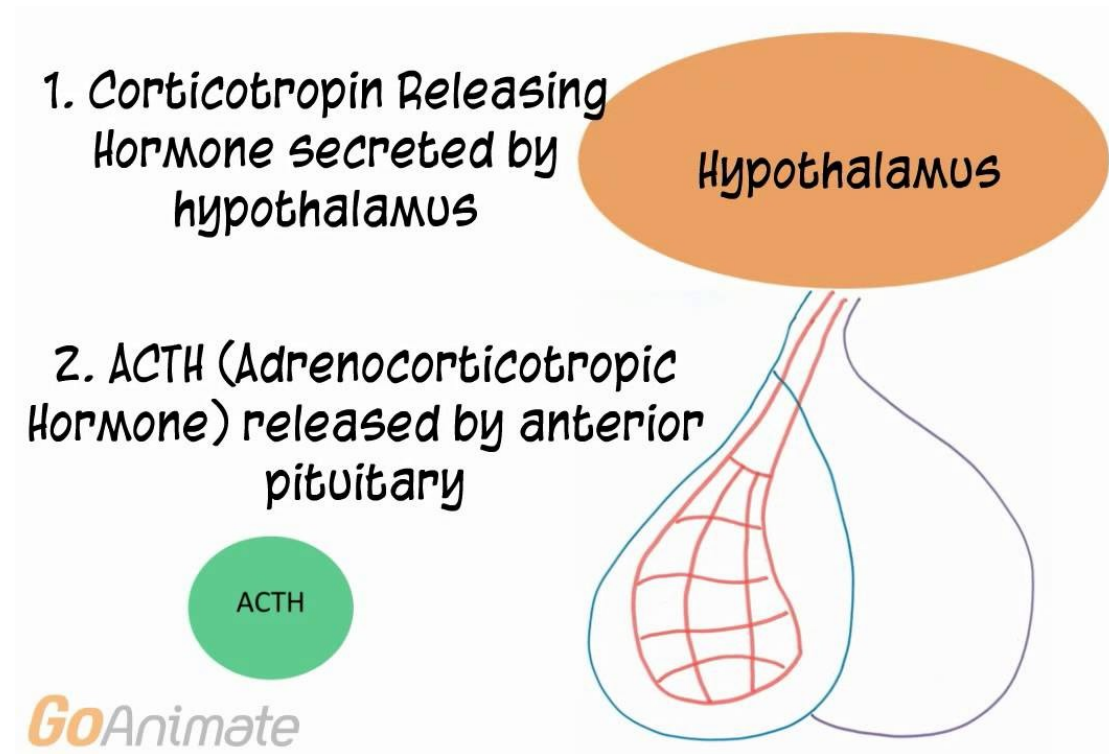


From Thibodeau GA, Patton KT: *The human body in health and disease*, ed 4, St Louis, 2005, Mosby.

ACTH - adrenocorticotrophic hormone

Target: Adrenal Glands

- Released in response to ANY stress
- ADRENAL GLAND releases:
- Corticosteroid – Glucocorticoid, cortisol (Steroid)
 - Increases BG, shuts off immune response, anti inflammatory, breaks down muscles.
- Adrenal gland also releases ANDROGENS/Testosterone
 - Promotes bone density/secondary sexual characteristics.



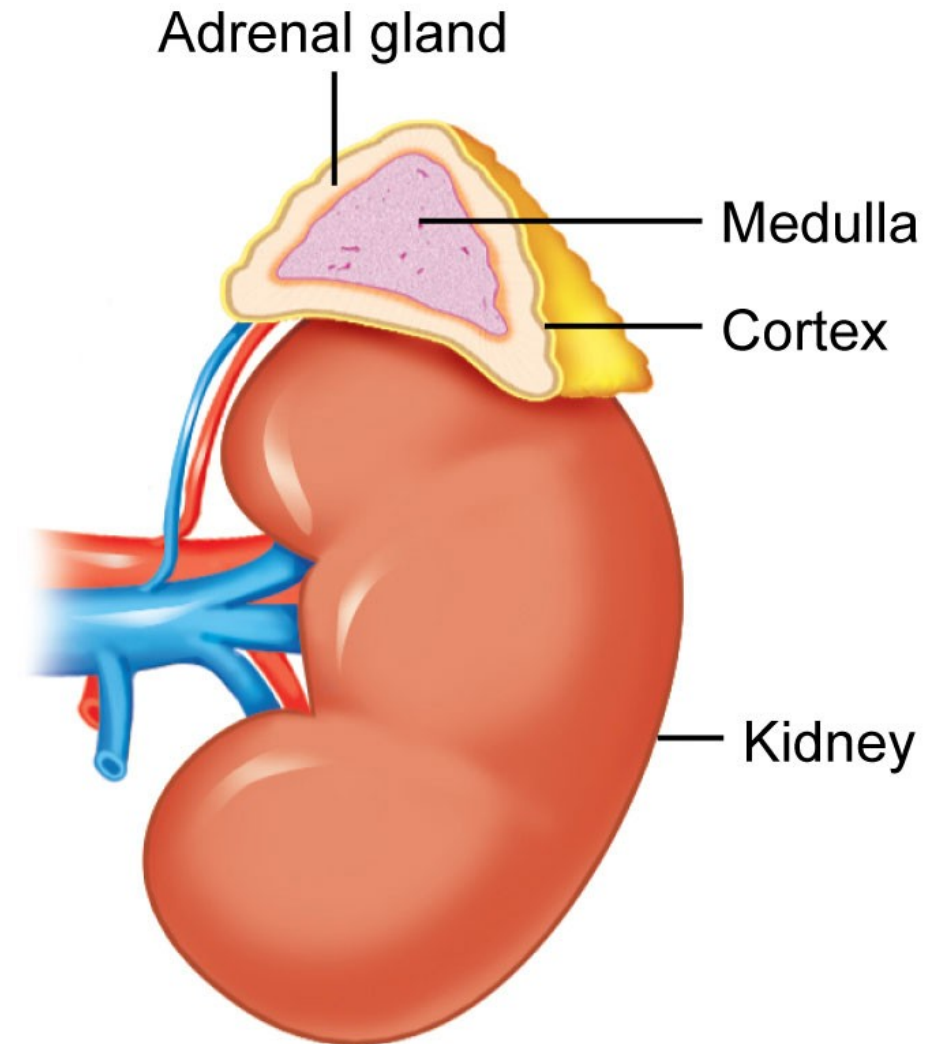
Other Adrenal Gland hormones

Aldosterone:

- Due to Low BP, Low blood Na^+ or high blood K^+
- Causes the kidney to absorb Na^+ (water follows)
 - Raises blood volume – BP goes up!
- Causes Kidney to PEE K^+

Epi and NorEpi:

- From Medulla DUE to sympathetic stimulation - fight or flight
 - Muscles work harder
 - Fats are released for energy
 - Glycogen released from liver
 - Heart beats faster and stronger



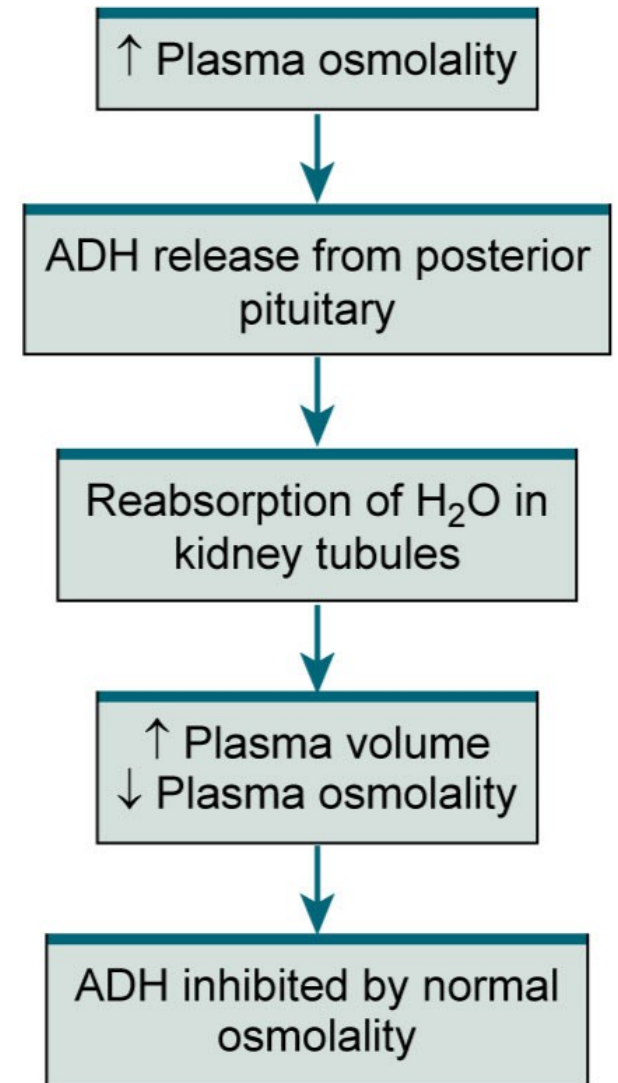
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ADH ANTIDIURETIC HORMONE -

from POST PIT

- released due to: LOW Fluid volume (thick blood) and/or Low blood pressure.
- Tells the kidneys to **reabsorb WATER** (anti-PEE)

ANP – Heart Hormone causes opposite impact as ADH and Aldosterone – released when TOO much volume = decrease in BP



OTHERS FROM THE ANT PIT: - Matching

A- Growth Hormone

1- Causes release of estrogen in females and sperm production/maturation in males.

B- Prolactin

2- Promotes tissue growth and repair

C - FSH

3- Causes milk production

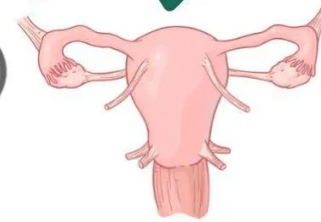
D - LH

4- Causes ovulation in females. Causes males to release testosterone.

Oxytocin = Uterine contractions during childbirth

OXYTOCIN

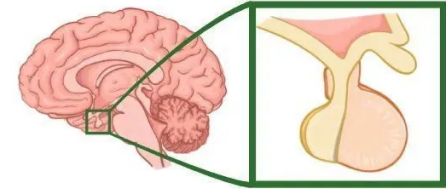
* PEPTIDE HORMONE released by POSTERIOR PITUITARY



UTERINE MUSCLE CONTRACTION



MILK LET-DOWN REFLEX



Pancreas: Insulin and Glucagon are **opposites!**

- ✓ **INSULIN:** Released by β cells
 - **Bolus** – when blood glucose is high or increasing
 - **Low**ers blood glucose by allowing it to enter cells or move into storage as glycogen (liver) or become triglyceride
 - ✓ **Glucagon:** Released by Alpha cells when blood glucose is low
 - **Increases** blood glucose by breaking down glycogen in muscles, gluconeogenesis in Liver, and breaks down triglycerides.
- ❖ Work together to maintain Blood Sugars between **70-120 mg/dL**

Practice Question

Blood glucose problems are most likely associated with which?

- A – adrenal cortex
- B – anterior pituitary
- C – pancreas
- D – thyroid gland

Practice Question

Kelly, a female, has been noticing excessive hair growth and an increase in her muscle mass. Which could be the reason behind these symptoms?

- A – the testes have begun producing androgens
- B – the adrenal cortex is producing excess androgens
- C – the ovaries are producing androgens
- D – the leptin secretion is low

Practice Question

Peter is not interested in eating. He says he's full. What hormone is too high?

- A – Glucagon
- B – Leptin
- C – Insulin
- D – Thyroxine

Where does this hormone come from?
What is its target?

Reminders

DUE WEDNESDAY: 1st discussion post

DUE SUNDAY:

- Pre-lesson checkpoint
- Module 8 quiz (Respondus)
- Peer response in discussion
- Lab quiz (Respondus)

You can avoid feeling overwhelmed on Sunday by completing something each day – this also leads to better retention of knowledge.

Struggling?

Let's meet – send me an email and we can arrange a meeting time!